Introduction

A funnel plot is the canonical visualisation for diagnosing publication bias in meta-analysis. It plots each study's effect size against its precision (or its standard error), forming an inverted-funnel shape under the null of no bias: large precise studies cluster near the pooled estimate at the narrow top of the funnel, smaller imprecise studies scatter wider at the base. Asymmetry of the funnel — typically a missing wedge of small null or negative studies — suggests that studies have been suppressed for non-statistical reasons, often because the field's preferred direction is positive and significant.

Prerequisites

A working understanding of meta-analysis, the relationship between sample size and standard error, and the difference between heterogeneity and publication bias as competing explanations for asymmetry.

Theory

A funnel plot has axes:

- **x**: effect size (log odds ratio, log hazard ratio, standardised mean difference, etc.).
- **y**: standard error (with smaller SEs at the top, so the funnel narrows upward) or precision 1/SE.

Under the null of no publication bias and no small-study effects, studies fall symmetrically inside a triangular pseudo-confidence funnel centred on the pooled estimate. Asymmetry — most commonly an absence of small studies on one side — is the visual cue for bias. Formal tests quantify it: Egger's regression of the standardised effect on its precision is the most widely used; Begg–Mazumdar's rank correlation is more robust to sparse data. Trim-and-fill (`metafor::trimfill()`) imputes missing studies to estimate a bias-adjusted effect.

Assumptions

A meta-analysis with at least 10 studies (fewer makes asymmetry impossible to judge), study-level effects and standard errors on a comparable scale, and the working assumption that no other sources (e.g. true heterogeneity correlated with study size) explain the asymmetry.

R Implementation

```r
library(metafor)

# Meta-analysis of 10 hypothetical studies
dat <- data.frame(
  yi = c(-0.1, -0.3, -0.5, -0.4, -0.2, -0.6, -0.7, -0.3, -0.9, -1.2),
  vi = c(0.04, 0.03, 0.08, 0.05, 0.02, 0.09, 0.07, 0.06, 0.10, 0.12)
)
res <- rma(yi = yi, vi = vi, data = dat, method = "REML")
funnel(res, main = "Funnel plot of 10 studies")

# Egger test
regtest(res, model = "lm")
```

Output & Results

A funnel plot with each study marked at its (y_i, SE_i) position, a vertical line at the pooled estimate, and pseudo-95 % confidence funnel boundaries that widen as standard error increases. `regtest()` returns Egger's test slope and p-value; a significant slope is conventional evidence of asymmetry.

Interpretation

A reporting sentence: "The funnel plot of 10 studies showed visible asymmetry (Egger's regression slope -2.1, $p = 0.04$); trim-and-fill imputed three additional studies on the right and shifted the pooled estimate from -0.55 (95 % CI -0.78 to -0.32) to -0.42 (95 % CI -0.66 to -0.17), suggesting that publication bias inflates the apparent effect by roughly 25 %." Always pair the visual with a formal test; a single test rarely provides strong evidence on its own.

Practical Tips

- Require at least 10 studies before drawing conclusions from a funnel plot; fewer studies cannot support a meaningful asymmetry test.
- True between-study heterogeneity correlated with study size (smaller studies in different populations) can mimic publication-bias asymmetry; consider both explanations.
- `metafor::trimfill()` adjusts for detected asymmetry but is not a cure — it is one of several sensitivity analyses to report.
- Contour-enhanced funnel plots (`funnel(res, level = c(90, 95, 99))`) overlay significance contours, helping to distinguish bias from heterogeneity.
- For binary outcomes, plot on the log-OR or log-RR scale; raw OR axes distort the symmetry under the null.
- Custom `ggplot2` funnel plots are easy to build (`geom_point()` plus `geom_segment()` for the funnel boundaries) when journal style requires consistent typography across figures.

R Packages Used

`metafor` for `rma()`, `funnel()`, `regtest()`, and `trimfill()`; `meta` for `metabias()` and alternative funnel implementations; `ggplot2` for hand-built funnel plots when the default base-graphics output does not match a publication's style; `dmetar` for additional bias diagnostics.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**
- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- ggplot2 grammar and multi-panel layouts